

PYTHON STRING METHODS

The Essential Guide to Text Manipulation

01. Standardizing Case

Techniques for normalizing user input and formatting display text.

| Changing Case

lower() & upper()

Ideal for case-insensitive comparisons and standardized storage.

```
msg = "Python 3"  
msg.lower() # "python 3"  
msg.upper() # "PYTHON 3"
```

title() & capitalize()

Perfect for formatting names, headings, and formal text outputs.

```
name = "john doe"  
name.title() # "John Doe"  
name.capitalize() # "John doe"
```

02. Cleaning & Trimming

Handling whitespace and unwanted characters in raw data.

| The strip() Family

`strip()` removes whitespace from both ends, while `rstrip()` and `lstrip()` target specific sides.

```
data = " raw input "  
data.strip() # "raw input"
```

Modern Python additions allow for clean prefix/suffix removal:

```
file = "data_v1.csv"  
file.removesuffix(".csv") # "data_v1"
```

 Minimalist cleaning icon

03. Searching & Inspecting

Discovering patterns and validating string content structure.

Start, End, and Find

Use `startswith()` and `endswith()` for logical checks on protocols, domains, or filetypes.

```
url = "https://example.com"  
url.startswith("https") # True
```

`find()` is the safer alternative to `index`, returning -1 if the search fails.

```
txt = "Learn Python"  
txt.find("Java") # -1
```



04. Modifying & Parsing

Splitting large text blocks into lists and reassembling them.

| Replace & Split

replace(old, new)

Swaps occurrences of a substring with a target string.

```
s = "I like Java"
s.replace("Java", "Python")
# "I like Python"
```

split(sep)

Parses a string into a list based on a delimiter.

```
csv = "1,John,NYC"
csv.split(",")
# ["1", "John", "NYC"]
```

05. Data Validation

Quickly identifying the nature of character content.

| Validation Summary

Method	Condition for True	Example
<code>isdigit()</code>	Contains only numbers (0-9)	"123".isdigit() → True
<code>isalpha()</code>	Contains only letters (A-Z)	"Py".isalpha() → True
<code>isalnum()</code>	Contains letters OR numbers	"v2".isalnum() → True
<code>isspace()</code>	Contains only whitespace	" ".isspace() → True

Questions?

Thank you for your attention!

 `python_docs.strings.py`

| Image Sources



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